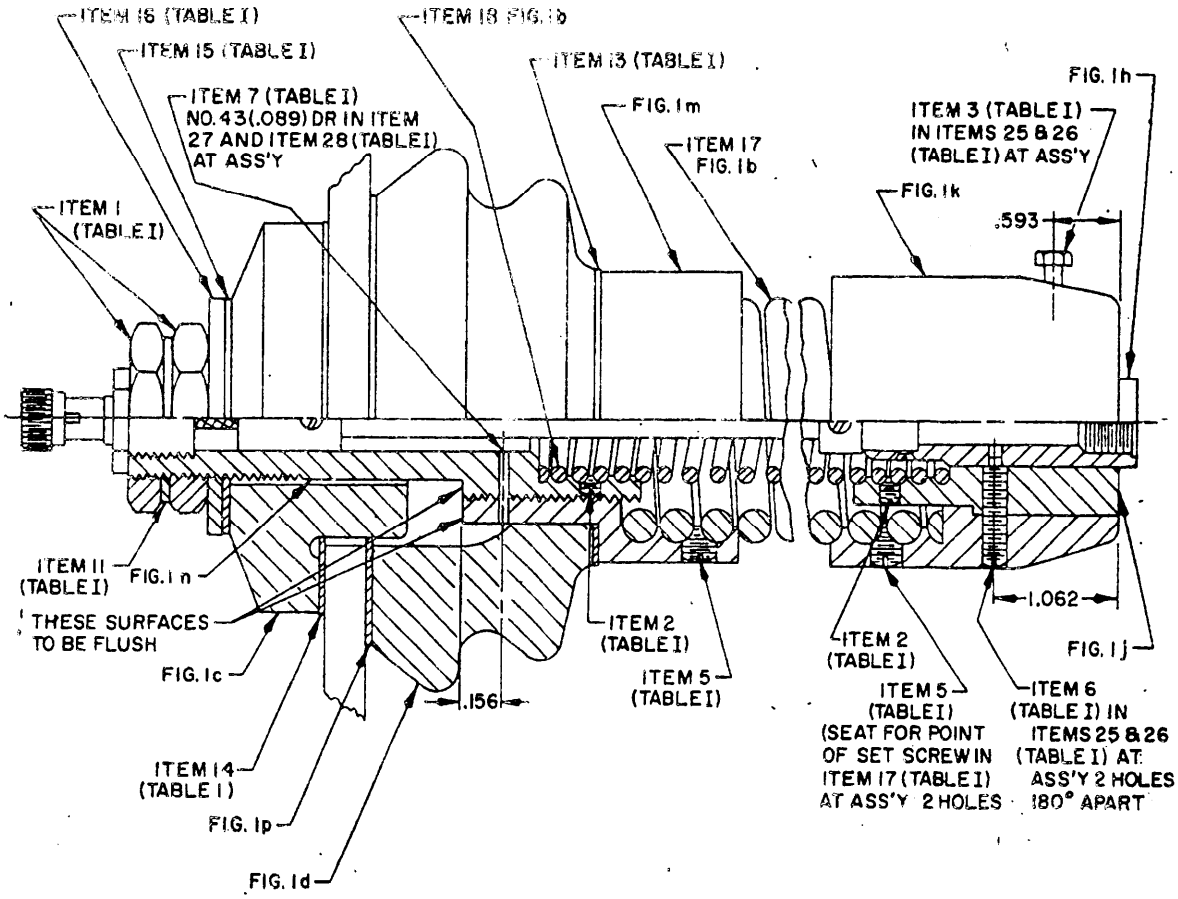
MILITARY SPECIFICATION SHEET

INSULATORS, FEEDTHRU
(STYLE 33)

The complete requirements for procuring the insulators described herein shall consist of this document and the latest issue of Specification MIL-I-23264.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.089	2.26
.156	3.96
.593	15.06
1.062	26.97

FIGURE 1a. Feedthru insulator assembly.

FIGURE 1. Style 33 feedthru insulator.

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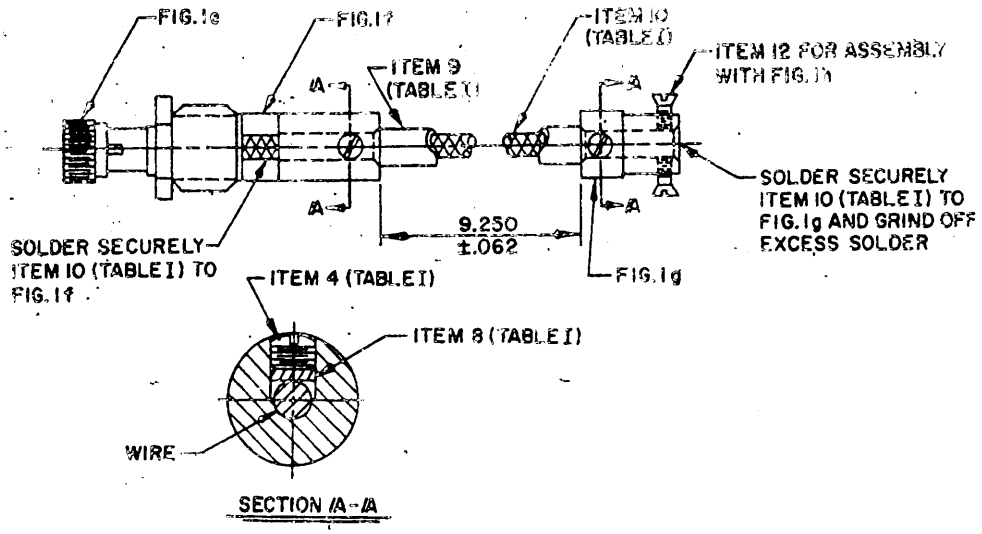
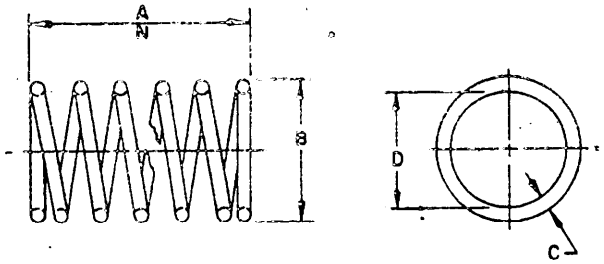


FIGURE 1a. Feedthru insulator assembly. - Continued.

INCHES	MM
.062	1.57
.125	3.18
.500	12.70
.750	19.05
1.125	28.58
1.500	38.10
2.000	50.80
3.250	82.55
9.250	234.95



Item No.	A	B	C	D	N (No. of turns)
17	5.500 (139.70) Free length	2.050 (52.07)	.312 (7.92)	1.425 (36.20)	18 Left hand
18	4.250 (107.95)	1.037 (26.34)	.146 No. 41 AMS & WGA	.741 (18.82)	29 Right hand

FIGURE 1b. Spring.
(Item 17 & 18, Table I)

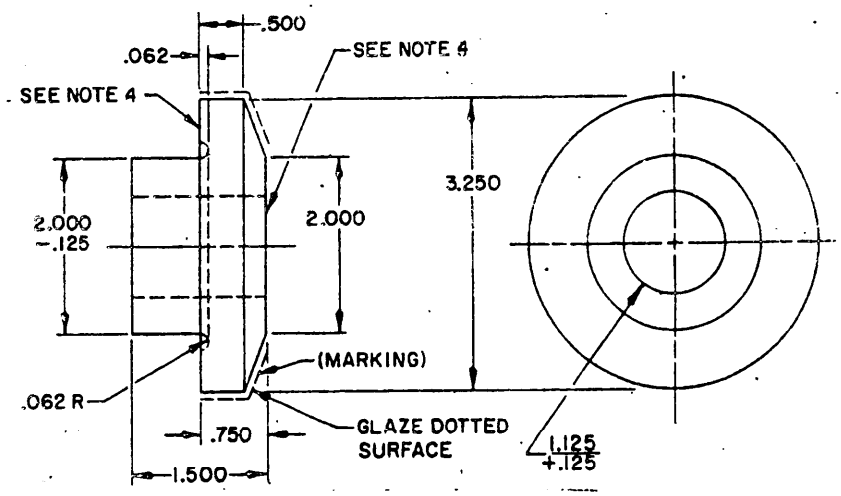
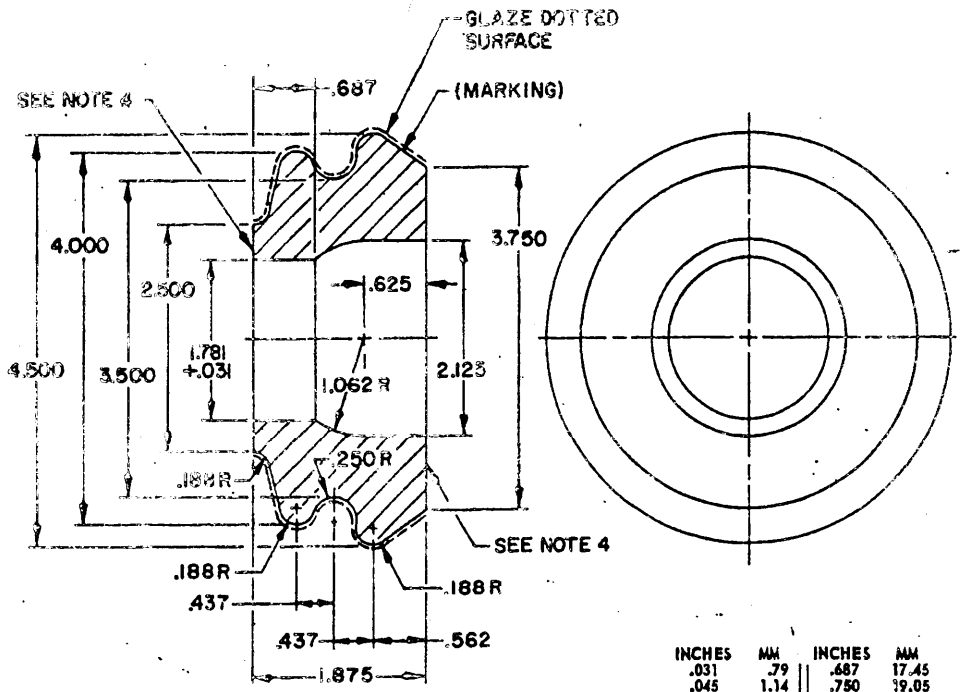


FIGURE 1c. Lower insulator.
(Item 19, Table I)

FIGURE 1. Style 33 feedthru insulator. - Continued.

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INCHES	MM	INCHES	MM
.031	.79	.687	17.45
.045	1.14	.750	19.05
.078	1.98	.937	23.80
.093	2.36	1.062	26.97
.140	3.56	1.781	45.24
.188	4.78	1.875	47.63
.250	6.35	2.125	53.98
.359	9.12	2.500	63.50
.437	11.10	3.500	88.90
.500	12.70	3.750	95.25
.562	14.27	4.000	101.60
.625	15.88	4.500	114.30

FIGURE 1d. Upper insulator.
(Item 20, Table I)

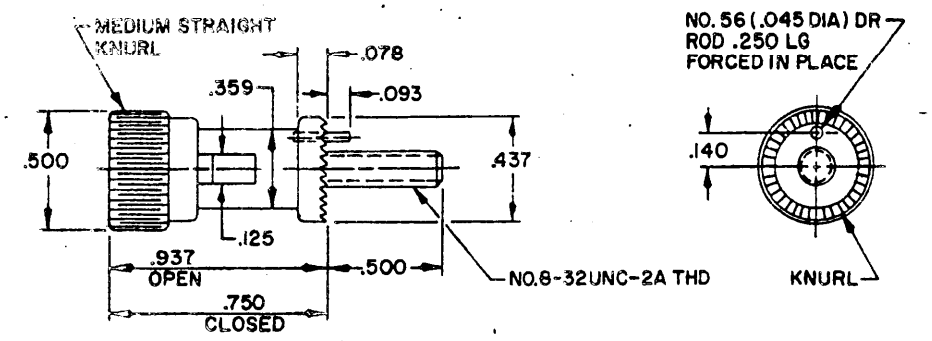


FIGURE 1e. Post.
(Item 21, Table I)

FIGURE 1. Style 33 feedthru insulator. - Continued.

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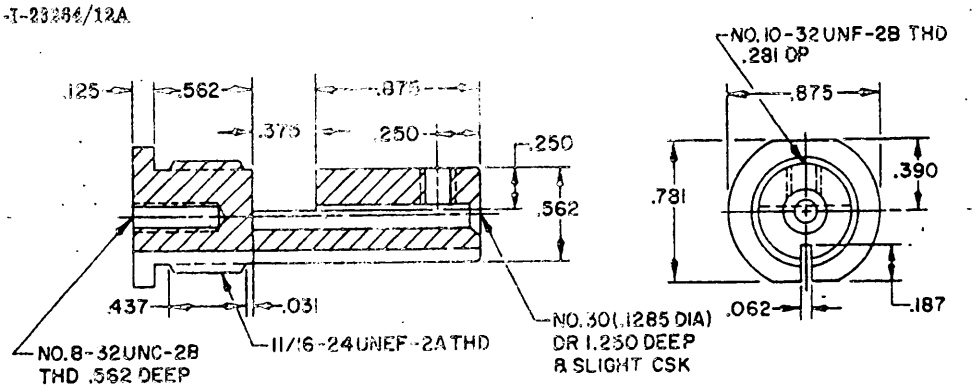
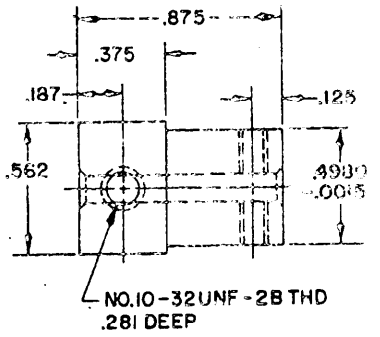


FIGURE 1f. Connector.
(Item 22, Table I)



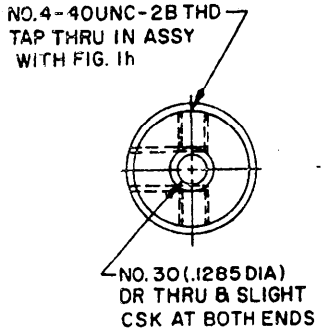


FIGURE 1g. Block.
(Item 23, Table I)

INCHES	MM
.001	.03
.003	.08
.031	.79
.062	1.57
.120	3.05
.125	3.18
.1285	3.26
.187	4.75
.2055	5.22
.250	6.35
.281	7.14
.375	9.53
.390	9.91
.422	10.72
.437	11.10
.4980	12.65
.500	12.70
.515	13.08
.562	14.27
.687	17.45
.743	18.87
.750	19.05
.781	19.84
.875	22.23
1.218	30.94
1.250	31.75
2.375	60.33

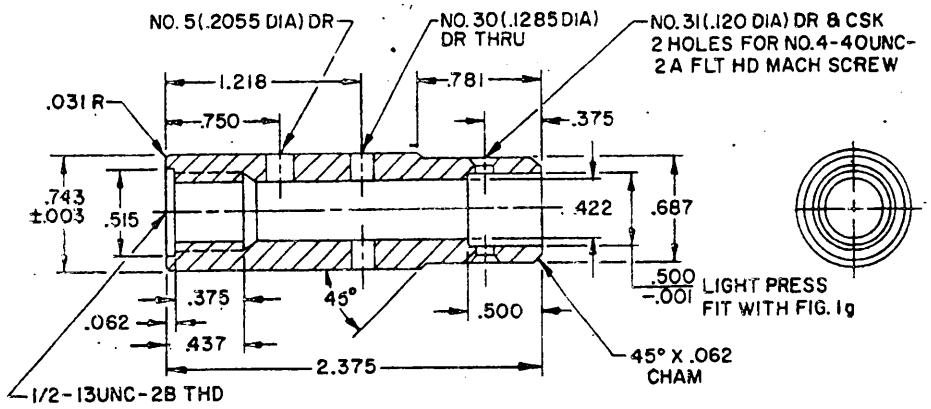


FIGURE 1h. Connector.
(Item 24, Table I)

FIGURE 1. Style 33 feedthru insulator. - Continued.

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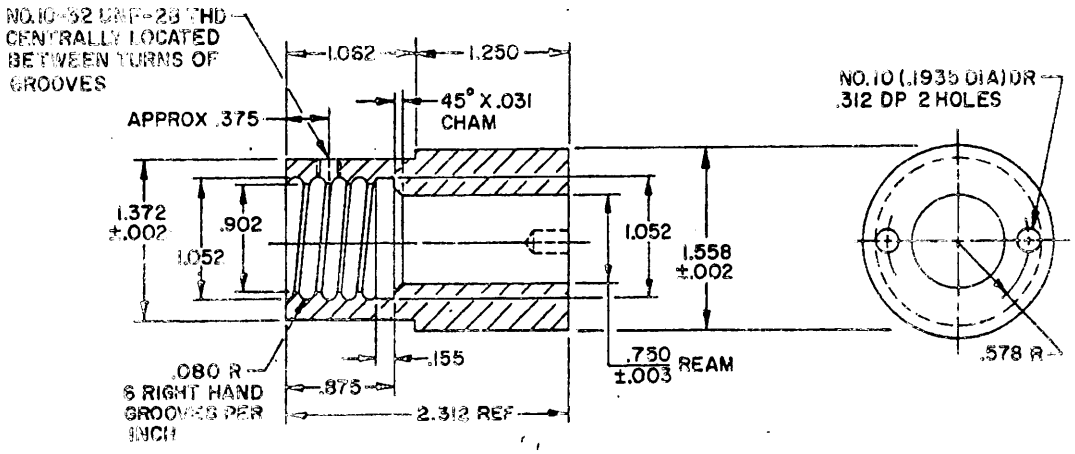


FIGURE 1j. Internal spring retainer.
(Item 25, Table I)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.1935	4.91	.902	22.91	1.562	39.75
.003	.08	.305	7.75	1.000	25.40	1.625	41.28
.010	.25	.312	7.92	1.052	26.72	1.812	46.02
.031	.79	.375	9.53	1.062	26.97	1.910	48.51
.080	2.03	.500	12.70	1.250	31.75	2.067	52.50
.125	3.18	.578	14.68	1.372	34.85	2.312	58.72
.155	3.94	.750	19.05	1.378	35.00	2.437	61.90
.158	4.01	.875	22.23	1.558	39.57	2.500	63.50

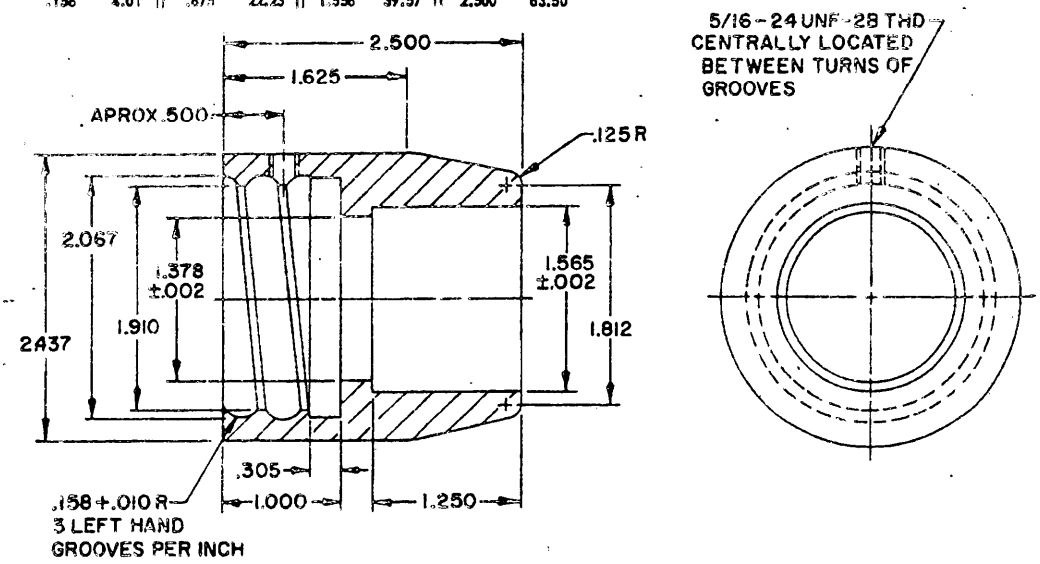


FIGURE 1k. External spring retainer.
(Item 26, Table I)

FIGURE 1. Style 33 feedthru insulator. - Continued.

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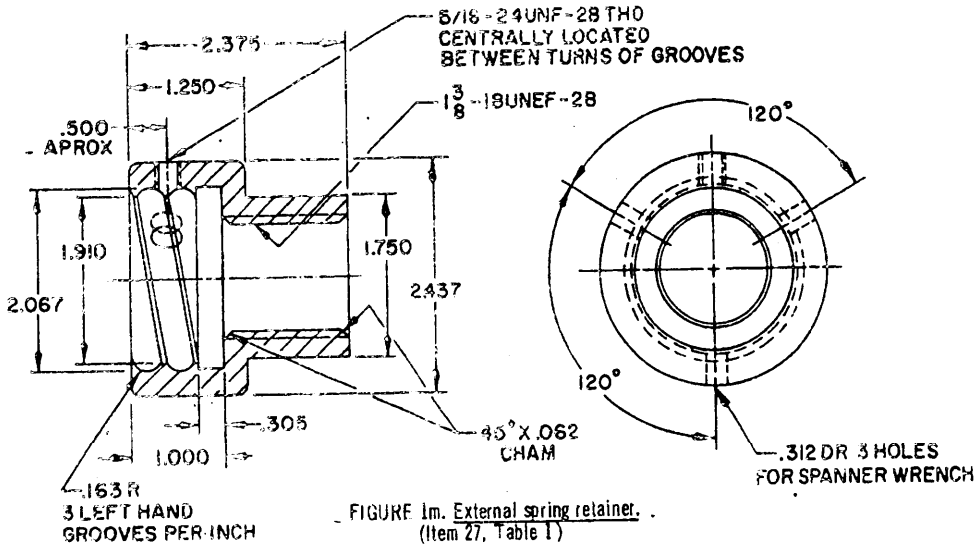


FIGURE 1m. External spring retainer.
(Item 27, Table I)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.031	.79	.312	7.92	1.000	25.40	2.067	52.50
.062	1.57	.375	9.53	1.052	26.72	2.375	60.33
.080	2.03	.500	12.70	1.250	31.75	2.437	61.90
.155	3.94	.575	14.61	1.500	38.10	2.500	63.50
.163	4.14	.875	22.23	1.750	44.45	2.875	73.03
.305	7.75	.905	22.99	1.910	48.51		

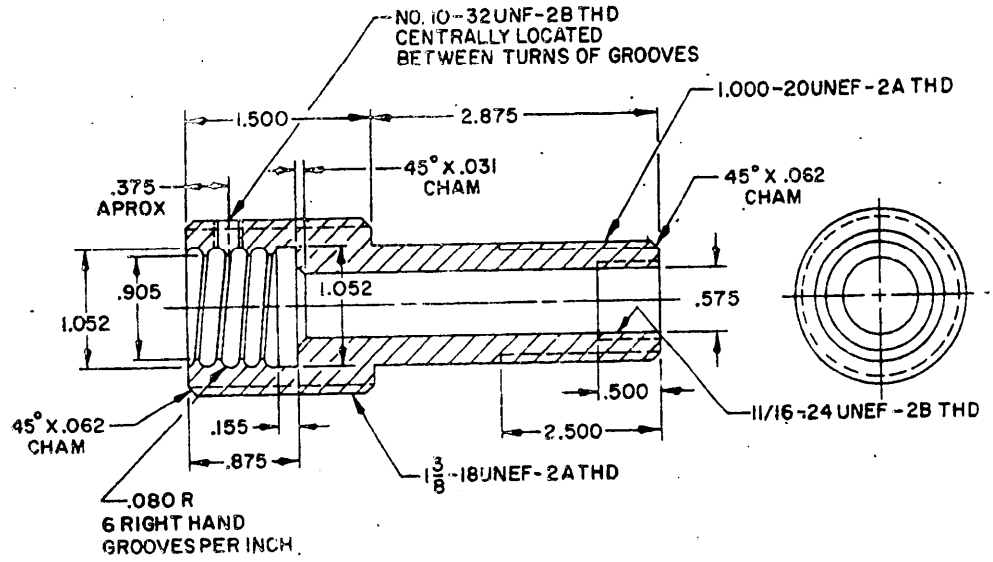
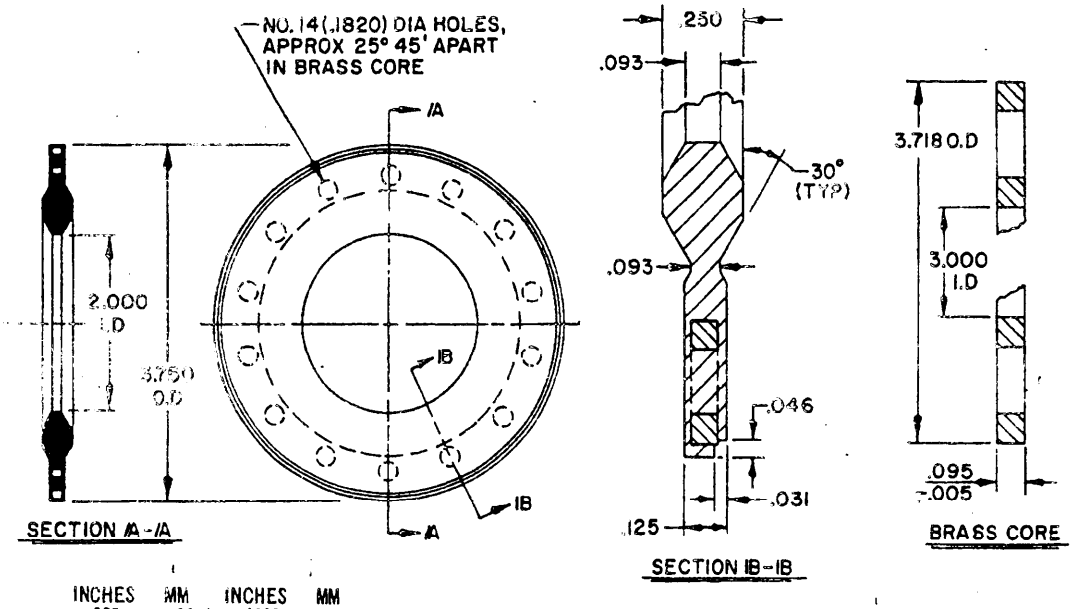


FIGURE 1n. Internal spring retainer.
(Item 28, Table I)

FIGURE 1. Style 33 feedthru insulator. - Continued.

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INCHES	MM	INCHES	MM
.005	.13	.1820	4.62
.031	.79	.250	6.35
.046	1.17	2.000	50.80
.093	2.36	3.000	76.20
.095	2.41	3.718	94.44
.125	3.18	3.750	95.25

FIGURE 1p. Gasket.
(Item 29, Table 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All exposed metal surfaces shall be painted in accordance with MIL-E-16400 except that a lustreless green enamel in accordance with TT-E-52T shall be used.
4. These surfaces to be smooth planes, perpendicular to axis of piece. Grind if necessary.

FIGURE 1. Style 33 feedthru insulator. - Continued.

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TABLE I. Item description.

Item No.	Description	Material	No. per assembly
1	Hexagon jam nut, 1-20UNEF-2B	Brass per QQ-B-637	2
2	Slotted-head, cone-point, set screw, 10-32 UNF-2A, .187 in. long	Corrosion resisting steel per Federal Standard No. 66, steel numbers 302 Rockwell "C" 35-43 or 420 Rockwell "C" 43-53	2
3	Slotted hexagon-head, cone-point, set screw, 10-32UNF-2A, 1 in. long		1
4	Flat-point, set screw, 10-32UNF-2A, .156 in. long		2
5	Slotted-head, cone-point, set screw, 5/16-24UNF-2A, .281 in. long		2
6	Full dog-point, slotted-head, set screw, 10-32UNF-2A, .93 in. long		2
7	Pin		1
8	Shoe	Brass per QQ-B-637	2
9	Tubing	Rubber, type 1, class 1, per MIL-C-1149	Length as required
10	Lead wire braid, .062 ID, 32 gage	Tinned copper per QQ-B-575	
11	Lockwasher per MS35338-110		1
12	Slotted flat-head, machine screw, 4-40UNC-2A, .250 in. long	Brass per QQ-B-637	2
13	Gasket, 2.437 OD, 1.750 ID, .062 THK, ±.005	Rubber, type 1, class 1, per MIL-C-1149	1
14	Gasket, 3.250 OD, 2.125 ID, .093 THK, ±.005		1
15	Gasket, 2.000 OD, 1.031 ID, .031 THK, ±.005	Fiber per MIL-F-1148	1
16	Washer, 2.000 OD, 1.031 ID, .125 THK, ±.005	Brass, composition 3, half hard, per QQ-B-637; silver plate .0002 min.	1
17	Spring, external	Steel	1
18	Spring, internal	Steel	1
19	Insulator, lower	Ceramic, grade 222 or better, per MIL-I-10	1
20	Insulator, upper		1

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TABLE I. Item description (continued).

Item No.	Description	Material	No. per assembly
21	Post	Brass, composition 2, half hard, per QQ-B-637	1
22	Connector		1
23	Block		1
24	Connector		1
25	Retainer, internal spring		1
26	Retainer, external spring		1
27	Retainer, external spring		1
28	Retainer, internal spring		1
29	Gasket	Rubber, type 1, class 1, per MIL-G-1149; brass core per QQ-B-637	1

TABLE II. Type designation.

Type designation	Dimensions
NL222B33-114	See figure 1

REQUIREMENTS:

Dimensions: See figure 1 and table I.

Tolerances:
Ceramic - Unless otherwise specified, the tolerances for ceramic parts shall be as specified in table IV of MIL-I-23264.
Metal - Unless otherwise specified, the tolerance for metal parts shall be ±.005 inch. Tolerance on anbles shall be ±1°.

Component parts:
Ceramic parts - Ceramic per MIL-I-10 shall be grade 222 or better.
Metal parts - Unless otherwise specified, all bearing and exposed metal surfaces shall be machine finished to 125/.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy - SH, EC
Air Force - 85
DSA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force

Preparing activity:
Navy - EC

(Project 5970-0161)

Revision letters are not used to indicate changes due to the extensiveness of the changes.

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